Online Learning Platform: ER Diagram (Chen notation)

LEGEND

- Entity
- Weak entity
- Attribute (attr)
- Primary key (key)
- Partial key (dashed underline) (pk)
- Derived attribute (attr)
- Multivalued attribute (attr)
- Relationship
- Identifying relationship
- Total participation (double line)
- Partial participation (single line)
- 1 / N = cardinality (1:1, 1:N)
- Composite attribute (splits into parts): attr → a, b
- Core
- Learning
- Commerce
- Engagement

INSTRUCTORS
- years_of_experience
- bio
- expertise
- rating
- total_earnings
- instructor_id

REVIEWS
- rating
- created_at
- comment
- review_id

USERS
- first_name
- last_name
- full_name
- user_id
- email
- password_hash
- role
- country
- profile_picture
- is_active
- created_at

COURSES
- price
- level
- description
- title
- language
- course_id
- duration_hours
- thumbnail_url
- status
- created_at

CATEGORIES
- name
- category_id
- description

SECTIONS
- section_order
- title
- section_id

LESSONS
- is_preview
- lesson_order
- duration_minutes
- lesson_id
- content_type
- content_url
- title

SUBMISSIONS
- submission_id
- feedback
- submitted_at
- file_url
- marks_obtained

ASSIGNMENTS
- assignment_id
- max_marks
- description
- title
- due_date

ORDERS
- ordered_at
- order_status
- total_amount
- order_id

ORDER_ITEMS
- price_at_purchase
- order_item_id

PAYMENTS
- payment_id
- paid_at
- amount
- payment_method
- transaction_id
- payment_status

ENROLLMENTS
- enrollment_id
- progress_percent
- status
- enrolled_at

CERTIFICATES
- certificate_id
- certificate_code
- issued_at

LESSON_PROGRESS
- progress_id
- completed_at
- last_watched_position
- is_completed

Relationships: is a, creates, writes, about, submits, receives, has, classifies, parent of, refers to, includes, places, contains, settles, for, enrolls, earns, records, tracks